note0403

On the identifiability of genetic marker effects in genomic models

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Often, the number of genetic marker covariates (m) included in genomic models exceeds the number of records (n). When the genomic model is parameterised in terms of genetic marker effects that are the focus of interest, given an $m \gg n$ setup, some form of regularisation is needed in order to obtain a solution for the m genetic marker effects using the n records available.

Consider a standard genomic model where y, an $n \times 1$ column vector of data has zero mean. The hierarchical structure of the model is as follows:

$$y \mid \beta, \sigma^2 \sim N\left(X\beta,\, I\sigma^2\right), \tag{1a}$$

$$\beta \mid \sigma^2_\beta \sim N\left(0,\, I\sigma^2_\beta\right). \tag{1b}$$

Regularisation is introduced by assigning a distribution to the vector β. The column vector of marker effects β, the focus of inference, is of dimension $m \times 1$ and in a first stage, it is assumed that the variance components σ^2 and σ^2_β are known so that β is the only unknown parameter. Matrix X is often rank deficient: $\text{rank}(X) = r$, $r \leq n$.

This note is in five parts: the first part describes how the data vector y contributes to the inference about the vector of marker effects β in this particular setup where the likelihood depends on β only through $X\beta$. The basic idea is to express the vector of marker effects as the sum of two orthogonal linear combinations of the elements of β: one that is identifiable and the other that is unidentifiable by the data. The second part examines the process of Bayesian learning in the genomic linear model (1) under two scenarios: one where variance components are known and the other where they are jointly inferred with the genetic marker effects β. The third, fourth and fifth parts discuss how identifiability of the vector of marker effects plays out in the context of the classical ridge regression, the lasso and the spike and slab model, respectively.

Preliminaries

1. Use will be made of the Moore-Penrose inverse (Searle, 1982) although this is not mandatory. Given a matrix X, the Moore-Penrose inverse X^+ is a *unique* matrix that satisfies the following criteria:

   - $XX^+X = X$
   - $X^+XX^+ = X^+$

- XX^+ is symmetric
- X^+X is symmetric

Given X, there are many matrices G that satisfy the first of these conditions. This is in contrast with the Moore-Penrose inverse that is unique.

Some of the results below require that the first and last condition hold. Although the Moore-Penrose inverse is used here, any (non-unique) generalised inverse G of matrix X satisfying these two conditions could also be used.

2. Singular value decomposition. In general, if X is an $(n \times m)$ matrix of rank r, then it can be written as

$$X = UDV', \tag{2}$$

where

- U is an $n \times r$ matrix whose columns are orthonormal left singular vectors of X;
- $D = \text{diag}(d_1, \ldots, d_r)$ is an $r \times r$ diagonal matrix whose diagonal elements satisfy $d_1 \geq d_2 \geq \cdots \geq d_r > 0$ and are the positive singular values of X;
- V is an $m \times r$ matrix whose columns are orthonormal right singular vectors of X.

The columns of U and V are orthonormal, that is, $U'U = I_r$, $V'V = I_r$, and unless $r = n$, $UU' \neq I_n$; rather, UU' is the orthogonal projector onto the column space of X. Similarly, unless $r = m$, $VV' \neq I_m$; instead, VV' is the orthogonal projector onto the row space of X.

3. Define the symmetric and idempotent $m \times m$ projection matrix P_R as

$$\begin{aligned} P_R &= X'(XX')^+X \\ &= X^+X \end{aligned} \tag{3}$$

where $(XX')^+$ is the Moore-Penrose inverse of XX' and X^+ is the Moore-Penrose inverse of X. Matrix P_R is the orthogonal projector of vectors in $\mathbb{R}^m$ onto the row space of matrix X (whose dimension is equal to r, the rank of X). This means that $\beta_R = P_R\beta$ lies in the row space of X, whereas $\beta_N = (I - P_R)\beta$ lies in the null space of X. Then,

$$\begin{aligned} X\beta_N &= X(I - P_R)\beta \\ &= X\beta - XX^+X\beta \\ &= 0 \end{aligned} \tag{4}$$

because $XX^+X = X$, due to the first Penrose identity.

Although β_R is an m-dimensional vector, it belongs to a subspace of dimension r, which is the dimension of the row space of matrix X. Therefore, only r independent coordinates are needed to specify β_R completely.

The proof of equality (3) is as follows. From the partitioning (2),

$$XX' = UDV'VDU' = UD^2U' \Rightarrow (XX')^+ = UD^{-2}U',$$

therefore

$$X'(XX')^+X = VDU'(UD^{-2}U')UDV' = VV'.$$

Inverting (2)

$$X^+ = VD^{-1}U' \Rightarrow X^+X = VD^{-1}U'UDV' = VV',$$

establishing the equality. In this expression, $D^{-1} = \text{diag}\left(\frac{1}{d_1}, \ldots, \frac{1}{d_r}\right)$, where $r = \text{rank}(X)$.

When the projection matrix is computed with the Moore-Penrose inverse, $P_R = P_R^2$ and $P_R' = P_R$; that is, P_R is idempotent and symmetric. However, the Moore–Penrose inverse is not required in the construction of these orthogonal projectors. Any generalized inverse satisfying $AGA = A$, with $A = XX'$ for P_R or $A = X'X$ for P_C, produces the same unique orthogonal projector.

Partitioning the vector of genetic marker effects

The vector β can be expressed as the sum of two orthogonal components

$$\begin{aligned} \beta &= P_R\beta + (I - P_R)\beta \\ &= \beta_R + \beta_N, \end{aligned} \tag{5}$$

where $\beta_R'\beta_N = 0$ (NOTE: $\beta_R'\beta_N = \beta' P_R'(I - P_R)\beta = \beta' 0 \beta = 0$, using $P_R = P_R'$, $P_R^2 = P_R$, both properties of the Moore-Penrose inverse). Then,

$$\begin{aligned} X\beta &= X\beta_R + X\beta_N \\ &= X\beta_R \end{aligned}$$

due to (4). Therefore the model $y = X\beta + e$ can be written as $y = X\beta_R + e$: data y contain information about the row space component, β_R, but not about the null space component, β_N. When the likelihood depends on β only through $X\beta$, the fraction of β that lies in the null space of X, $(I - P_R)\beta$, is not identifiable from y. The crucial point is that β_R is identifiable because it is uniquely determined by $X\beta$: knowing β_R determines $X\beta$, and knowing $X\beta$, determines β_R.

When the rank of X is $r = m$, $VV' = I_m$. In this case $P_R = I_m$, $(I_m - P_R) = 0$, and $\beta_N = 0$, indicating that the null space has disappeared; the entire vector β is identifiable from the data. Then

$$P_R\beta = \beta.$$

Parametrising the genomic model in terms of genomic values

Assume that instead of parametrising the genomic model in terms of marker effects β, the model takes the form:

$$y \mid g, \sigma^2 \sim N\big(g,\, I\sigma^2\big)\,, \tag{6a}$$

$$g \mid \sigma^2_G \sim N\big(0,\, G\sigma^2_G\big) \tag{6b}$$

where $G = \frac{XX'}{m}$ and $\sigma^2_G = m\sigma^2_\beta$. The covariance matrix $G\sigma^2_G$ is positive semidefinite (possibly singular), so g has a singular multivariate normal distribution whose support is the column space of X, whenever $\text{rank}(X) < n$, $m > n$. Despite the rank deficiency of X, under this parametrisation, every admissible value of the parameter g is identifiable, as shown below.

Define $P_C = XX^+$, the symmetric and idempotent $n \times n$ matrix that projects vectors in $\mathbb{R}^n$ onto the column space of X. Notice that this is in contrast with the $m \times m$ matrix P_R, that projects vectors in $\mathbb{R}^m$ onto the row space of X. Then, with $g = X\beta$,

$$\begin{aligned} P_C\, g &= XX^+X\beta \\ &= X\beta = g \end{aligned}$$

because $XX^+X = X$, due to the first Penrose identity, and $(I_m - P_C)g = (X\beta - XX^+X\beta)$ $= 0$. Every admissible parameter g lies in the column space of X; thus the parameter space coincides with the identifiable parameter space that has dimension r, the rank of X, and the projector P_C is the identity operator. There is no unidentifiable component of g, in contrast with the partitioning in (5): for β, the parameter space is $\mathbb{R}^m$ but only the projection onto the row space of X is identifiable. Thus, despite the fact that β_R and g have m and n elements, respectively, they both belong to a space of dimension r. Furthermore,

$$g = X\beta_R, \qquad \beta_R = X^+g,$$

therefore the mapping between them is one-to-one (Notice: $X^+g = X^+X\beta$; $X^+X\beta = P_R\beta = \beta_R$). Consequently, the data contain the same statistical information about β_R as about g, since these parameters are related by the one-to-one transformation. Or, to put it differently, the m elements of β_R and the n elements of g are simply two different representations of the same r identifiable degrees of freedom.

The second part of this note examines how identifiability of β plays out when matrix X of the genomic model (1) is also rank deficient but variance components σ^2_β and σ^2 are unknown and updated by the data. Before discussing the subject an intuitive explanation of the concept of "identifiability" is provided below, using some of the ideas introduced so far.

A little on identifiability

The previous discussion indicated that matrix P_R is the orthogonal projector of β, $\beta \in \mathbb{R}^m$, onto the row space of the rank deficient $n \times m$ matrix X. Then $\beta_R = P_R\beta$ lies in the row

space of X. An explanation is provided as to why β_R is identifiable under the model $y = X\beta + e$, provided that the probability distribution of y identifies $X\beta$.

Suppose that two different parameter vectors, $\beta^{(1)}$ and $\beta^{(2)}$, satisfy

$$X\beta^{(1)} = X\beta^{(2)}.$$

Then,

$$X(\beta^{(1)} - \beta^{(2)}) = 0.$$

This means that the difference between the two parameter vectors $d = \beta^{(1)} - \beta^{(2)}$ belongs to the null space of X. Recall the definition of null space: it consists of all vectors that are transformed into the zero vector when multiplied by X:

$$\text{Null}(X) = \{v \in \mathbb{R}^m : Xv = 0\}$$

therefore

$$d \in \text{Null}(X).$$

Since the null space and the row space are orthogonal, $P_R d = 0$ for every $d \in \text{Null}(X)$ (ie, for every $d \in \text{Null}(X)$, $Xd = 0$. Since $P_R = X^+X$, $P_R d = X^+Xd = 0$). Consequently, if

$$\beta^{(1)} = \beta^{(2)} + d,$$

then,

$$\begin{aligned} P_R\beta^{(1)} &= P_R\beta^{(2)} + P_R d \\ &= P_R\beta^{(2)}. \end{aligned}$$

Therefore

$$\beta_R^{(1)} = \beta_R^{(2)}.$$

Although two parameter vectors may differ, if they produce the same $X\beta$, their difference lies entirely in the null space. Projecting them onto the row space eliminates that difference. Hence, all parameter vectors producing the same linear predictor $X\beta$ have exactly the same row space component β_R. This is the fundamental reason why β_R is uniquely determined by $X\beta$, and therefore statistically identifiable whenever $X\beta$ itself is identifiable.

Bayesian learning: updating genetic marker effects

The question addressed is whether, given a hierarchical Bayesian model and a body of data, Bayesian inference leads to learning about all components of vector β. This is investigated by comparing the prior and posterior distributions of the marker effects. If the posterior distribution of a component is identical to its prior distribution, the data provide no information about that component beyond that already contained in the prior, and no Bayesian learning has occurred for that component.

Variance components known

Two cases are discussed. The first assumes that the prior distribution $\beta \mid \sigma^2_\beta \sim N(0, I\sigma^2_\beta)$; the second assigns a correlation structure via the parameter Σ: $\beta \mid \sigma^2_\beta \sim N(0, \Sigma\sigma^2_\beta)$, where Σ is a known, symmetric, positive definite matrix.

1. Consider again the standard genomic model that is described by the following hierarchy:

$$y \mid \beta, \sigma^2 \sim N\big(X\beta,\, I\sigma^2\big)\,, \tag{7a}$$

$$\beta \mid \sigma^2_\beta \sim N\big(0,\, I\sigma^2_\beta\big)\,. \tag{7b}$$

The vector y contains the n phenotypic observations, X is the observed $(n \times m)$ marker matrix with $m > n$ and rank $r < \min(n, m)$, and β is the unobserved $(m \times 1)$ vector of marker effects. The variance components σ^2 and σ^2_β are assumed known.

As indicated in the first part of this note, the vector of genetic marker effects can be partitioned into a component that is identifiable from the data, β_R, and an orthogonal component that is not identifiable from the data, β_N.

From Bayes theorem,

$$p(\beta_R, \beta_N \mid y, \sigma^2, \sigma^2_\beta) = \frac{p(y \mid \beta_R, \beta_N, \sigma^2)p(\beta_R, \beta_N \mid \sigma^2_\beta)}{p(y \mid \sigma^2, \sigma^2_\beta)}.$$

First, since $X\beta_N = 0$, the likelihood depends only on β_R,

$$p(y \mid \beta_R, \beta_N, \sigma^2) = p(y \mid \beta_R, \sigma^2).$$

Second, since $p(\beta_R, \beta_N \mid \sigma^2_\beta) = p(\beta_R \mid \sigma^2_\beta)p(\beta_N \mid \sigma^2_\beta)$ (see NOTE, below). Substituting into the expression of Bayes theorem,

$$p(\beta_R, \beta_N \mid y, \sigma^2, \sigma^2_\beta) = \frac{p(y \mid \beta_R, \sigma^2)p(\beta_R, \mid \sigma^2_\beta)}{p(y \mid \sigma^2, \sigma^2_\beta)}p(\beta_N \mid \sigma^2_\beta).$$

The first factor is recognised as $p(\beta_R \mid y, \sigma^2, \sigma^2_\beta)$, therefore

$$p(\beta_R, \beta_N \mid y, \sigma^2, \sigma^2_\beta) = p(\beta_R \mid y, \sigma^2, \sigma^2_\beta)p(\beta_N \mid \sigma^2_\beta). \tag{8}$$

Integrating over β_R yields

$$\begin{aligned} p(\beta_N \mid y, \sigma^2, \sigma^2_\beta) &= p(\beta_N \mid \sigma^2_\beta) \int p(\beta_R \mid y, \sigma^2, \sigma^2_\beta)\, d\beta_R \\ &= p(\beta_N \mid \sigma^2_\beta) \end{aligned} \tag{9}$$

because the second term integrates to 1. The equality of the posterior and prior distributions of β_N indicates lack of Bayesian learning under the Gaussian prior of

β with independent marker effects and a common variance. A further observation is that, from the joint posterior factorisation of (β_R, β_N) in (8),

$$p(\beta_N \mid \beta_R, y, \sigma^2, \sigma^2_\beta) = p(\beta_N \mid \sigma^2_\beta).$$

Thus, β_R and β_N remain independent under the posterior distribution.
Conclusion: The likelihood updates the prior distribution of β_R, while leaving the distribution of β_N unchanged. The reason is not simply that $X\beta_N = 0$, but that the Gaussian prior of β with independent marker effects and a common variance, additionally, makes β_R and β_N independent.

NOTE

With $\beta|\sigma^2_\beta \sim N(0, I_m\sigma^2_\beta)$ and $\beta_R = P_R\beta$, $\beta_N = (I_m - P_R)\beta$,

$$\begin{aligned} Cov(\beta_R, \beta'_N \mid \sigma^2_\beta) &= Cov(P_R \operatorname{Var}(\beta \mid \sigma^2_\beta)(I_m - P_R)'|\sigma^2_\beta) \\ &= P_R(I_m - P_R)\sigma^2_\beta = 0 \end{aligned}$$

since P_R is symmetric and idempotent, a property of the Moore-Penrose inverse. Therefore, given the Gaussian prior of β with independent marker effects and a common variance, $p\left(\beta|\sigma^2_\beta\right) = p\left(\beta_R|\sigma^2_\beta\right) p\left(\beta_N|\sigma^2_\beta\right)$.

2. In the second case, the hierarchical genomic model is

$$y \mid \beta, \sigma^2 \sim N\left(X\beta,\, I\sigma^2\right), \tag{10a}$$
$$\beta \mid \sigma^2_\beta \sim N\left(0,\, \Sigma\sigma^2_\beta\right). \tag{10b}$$

Begin by writing the posterior distribution,

$$p(\beta_R, \beta_N \mid y, \sigma^2, \Sigma\sigma^2_\beta) \propto p(y \mid \beta_R, \sigma^2)p(\beta_R, \beta_N \mid \Sigma\sigma^2_\beta).$$

Integrating over β_R results in

$$p(\beta_N \mid y, \sigma^2, \Sigma\sigma^2_\beta) \propto \int p(y \mid \beta_R, \sigma^2)p(\beta_R, \beta_N \mid \Sigma\sigma^2_\beta)\, d\beta_R.$$

Factorising the prior yields

$$\begin{aligned} p(\beta_N \mid y, \sigma^2, \Sigma\sigma^2_\beta) \propto\, & p(\beta_N \mid \Sigma\sigma^2_\beta) \\ & \times \int p(y \mid \beta_R, \sigma^2)p(\beta_R \mid \beta_N, \Sigma\sigma^2_\beta)\, d\beta_R. \end{aligned} \tag{11}$$

The integral represents the likelihood of y conditional on β_N, after integrating out β_R. The posterior distribution of β_N differs from its prior distribution, indicating Bayesian learning given the model specified in (10).

The likelihood $p(y \mid \beta_R, \beta_N, \sigma^2)$ contains no direct information about β_N, because it depends only on β_R. However, after integrating over β_R, the resulting distribution $p(y \mid \beta_N, \sigma^2, \Sigma\sigma^2_\beta)$ may depend on β_N through the prior dependence between β_R and β_N induced by the prior covariance structure (see the NOTE below). This dependence permits indirect Bayesian learning about β_N, even though β_N remains non-identifiable through the original likelihood.

What form does expression (11) take, assuming prior independence of β_R and β_N such that $p(\beta_R \mid \beta_N, \Sigma\sigma^2_\beta) = p(\beta_R \mid \Sigma\sigma^2_\beta)$? If this equality holds, substituting in (11) results in

$$p(\beta_N \mid y, \sigma^2, \Sigma\sigma^2_\beta) \propto p(\beta_N \mid \Sigma\sigma^2_\beta) \times \int p(y \mid \beta_R, \sigma^2) p(\beta_R \mid \Sigma\sigma^2_\beta)\, d\beta_R. \tag{12}$$

The integral is the marginal likelihood of y, $p(y \mid \sigma^2, \Sigma\sigma^2_\beta)$, which is a constant with respect to β_N. Therefore it can be absorbed into the proportionality constant yielding

$$p(\beta_N \mid y, \sigma^2, \Sigma\sigma^2_\beta) = C\, p(\beta_N \mid \Sigma\sigma^2_\beta)$$

where C does not depend on β_N. Integrating both sides over β_N and noting that both densities integrate to 1, gives $C = 1$ and

$$p(\beta_N \mid y, \sigma^2, \Sigma\sigma^2_\beta) = p(\beta_N \mid \Sigma\sigma^2_\beta),$$

recovering (9). Therefore, under the Gaussian genomic model with known variance components, absence of Bayesian learning about the non-identifiable component β_N occurs if and only if β_R and β_N are independent under the prior, and therefore $p(\beta_R \mid \beta_N, \Sigma\sigma^2_\beta) = p(\beta_R \mid \Sigma\sigma^2_\beta)$. This condition is automatically satisfied by a Gaussian prior of β with independent marker effects and a common variance, but can also hold for more general covariance structures. As a simple example, consider the following:

$$\Sigma = aP_R + b(I - P_R), \quad a, b > 0.$$

Then,

$$\begin{aligned} P_R\Sigma(I - P_R) &= P_R\left[aP_R + b(I - P_R)\right](I - P_R) \\ &= aP_R^2(I - P_R) + bP_R(I - P_R)^2 \\ &= 0. \end{aligned}$$

Thus, β_R and β_N are independent under the Gaussian prior, even though $\Sigma \neq I_m$ when $a \neq b$ (when $a = b$, $aP_R + a(I_m - P_R) = aI_m$). This covariance structure results in

$$\begin{aligned} \mathrm{Var}(\beta_R \mid \sigma^2_\beta) &= a\sigma^2_\beta P_R, \\ \mathrm{Var}(\beta_N \mid \sigma^2_\beta) &= b\sigma^2_\beta (I_m - P_R). \end{aligned}$$

An interesting consequence is that the individual marker effects β_i need not be independent, even though β_R and β_N are independent.

NOTE

Assume the prior takes the form $\beta \mid \sigma_\beta^2 \sim N(0, \Sigma\sigma_\beta^2)$, where Σ is a known, symmetric, positive definite matrix not necessarily equal to I_m. As shown before,

$$\begin{aligned}\beta_R &= P_R\beta,\\ \beta_N &= (I - P_R)\beta,\end{aligned}$$

where $P_R = X^+X$ is symmetric and idempotent. Then

$$\begin{aligned}\mathrm{Cov}(\beta_R, \beta_N' \mid \sigma_\beta^2) &= P_R \,\mathrm{Var}(\beta \mid \sigma_\beta^2)(I_m - P_R)'\\ &= \sigma_\beta^2 P_R\Sigma(I_m - P_R).\end{aligned}$$

Consequently, when $P_R\Sigma(I - P_R) \neq 0$, the prior does not factorise

$$p(\beta_R, \beta_N \mid \sigma_\beta^2) \neq p(\beta_R \mid \sigma_\beta^2)p(\beta_N \mid \sigma_\beta^2)$$

The likelihood still depends only on β_R, since $X\beta_N = 0$. Nevertheless, learning about β_R can now induce learning about β_N through their prior dependence.

Conclusion: When the variance components are known and the marker effects have an iid Gaussian prior, the orthogonal components β_R and β_N are uncorrelated and therefore independent. Consequently, their joint prior distribution factorises. When the marker effects have a more general Gaussian covariance structure, the two components remain orthogonal, but they are not necessarily independent. Their covariance is determined by $P_R\Sigma(I_m - P_R)$. If this matrix is nonzero, the prior does not factorise; if it is zero, independence and factorisation are preserved. The essential distinction is that orthogonality of the two projected vectors is a geometric property, whereas independence is a probabilistic property determined by the prior covariance structure.

Variance components unknown

Assume that σ_β^2 is unknown and updated from the data, and for simplicity, assume σ^2 is known. The posterior distribution of β_N takes the following form:

$$p\left(\beta_N|y, \sigma^2\right) = \int p\left(\beta_N|y, \sigma_\beta^2, \sigma^2\right) p\left(\sigma_\beta^2|y, \sigma_\beta^2\right) d\sigma_\beta^2.$$

Since $p\left(\beta_N|y, \sigma_\beta^2, \sigma^2\right) = p\left(\beta_N|\sigma_\beta^2\right)$, the posterior distribution becomes

$$p\left(\beta_N|y, \sigma^2\right) = \int p\left(\beta_N|\sigma_\beta^2\right) p\left(\sigma_\beta^2|y, \sigma^2\right) d\sigma_\beta^2.$$

This expression is different from

$$\int p\left(\beta_N|\sigma_\beta^2\right) p\left(\sigma_\beta^2\right) d\sigma_\beta^2$$

unless $p\left(\sigma^2_\beta|y,\sigma^2\right) = p\left(\sigma^2_\beta\right)$ which is usually not the case since the data updates σ^2_β. However, the mean of the posterior distribution of β_N remains unchanged relative to the prior mean. Indeed,

$$\int\int \beta_N\, p\left(\beta_N|\sigma^2_\beta\right) p\left(\sigma^2_\beta|y,\sigma^2\right) d\beta_N d\sigma^2_\beta =$$
$$\int\left[\int \beta_N\, p\left(\beta_N|\sigma^2_\beta\right) d\beta_N\right] p\left(\sigma^2_\beta|y,\sigma^2\right) d\sigma^2_\beta = 0$$

since the integral in square brackets corresponds to the prior mean.

Conclusion: when β and σ^2_β are inferred from a joint Bayesian analysis, the nonidentifiable component β_N of β undergoes indirect Bayesian learning through the posterior updating of the variance component σ^2_β. The likelihood remains completely flat in the β_N direction, so the identifiability structure of the model is unchanged: β_N is still not directly identifiable from the data. These results and those presented in the first part of this note, highlight the distinction between inference of genomic values $g = X\beta$ and inference of marker effects β. Genomic values are identifiable whereas only the β_R component of β is learned directly from the data. Consequently in genomic models where $m \gg n$, inference of individual marker effects is inherently more challenging than inference of genomic values.

How Does the Penalty Term in Ridge Regression Alter the Identifiability of β?

The first part of this note discussed identifiability of the parameter β of the linear regression model $y = X\beta + e$ when matrix X is rank deficient. It was indicated that the vector of genetic marker effects β can be partitioned into two orthogonal linear combinations: β_R that is identifiable by the data y, and β_N that is not (see page 3). The second part of this note shows how a Bayesian implementation deals with the rank deficiency of matrix X in order to infer β. When variance components are known and the only unknown parameter is β, given a Gaussian prior distribution for β of the form $N(0, I\sigma^2_\beta)$, the Bayesian analysis updates β_R via the data, but the information to infer the unidentifiable component β_N originates solely from the prior distribution. The posterior mode of β_N (and given the Gaussian assumption, the posterior mean) is equal to zero. This third part illustrates how this process unfolds in the context of ridge regression.

Referring to page 3, partition β into its two orthogonal components:

$$\beta = \beta_R + \beta_N. \tag{13}$$

The ridge coefficients are obtained as

$$\hat{\beta} \underset{\arg\min\ \beta}{=} (y - X\beta)'(y - X\beta) + \lambda\beta'\beta,$$

where here, the shrinkage parameter $\lambda > 0$. Using (13), the orthogonality of β_R and β_N, and the equality $X\beta = X\beta_R$, the cost function of the ridge regression is

$$J(\beta_R, \beta_N|\lambda) = (y - X\beta_R)'(y - X\beta_R) + \lambda\beta'_R\beta_R + \lambda\beta'_N\beta_N.$$

For any fixed value of β_R, the first two terms are constant with respect to β_N. Therefore

$$\hat{\beta}_N \underset{\arg\min \beta_N}{=} \lambda\, \beta_N' \beta_N.$$

Since $\lambda > 0$, $\lambda\, \beta_N' \beta_N \geq 0$, with equality only if $\beta_N = 0$. Thus,

$$\hat{\beta}_N = 0. \tag{14}$$

The remaining component $\hat{\beta}_R$ is obtained by minimising

$$(y - X\beta_R)'(y - X\beta_R) + \lambda\, \beta_R' \beta_R.$$

Conclusion: Ridge regression only estimates the component of β that is identifiable by the data: the data determine only β_R. The unidentifiable component β_N is set to zero because this minimises the cost function: the penalty determines β_N. The solution for β_N agrees with the Bayesian approach under the Gaussian prior for β: the posterior distribution of β_N is equal to its prior distribution, whose mean (and mode) is zero.

How Does the Penalty Term in Lasso Alter the Identifiability of β?

This section summarises the concept of identifiability of genetic marker effects β, when the matrix of observed marker effects X is rank deficient and lasso is used as the operational model. As before, partition vector β into its orthogonal components

$$\beta = \beta_R + \beta_N. \tag{15}$$

Then, since b_N belongs to the null space of X, $X\beta = X\beta_R$, and the loss function associated with lasso can be written as

$$J(\beta_R, \beta_N | \lambda) = (y - \beta_R)'(y - X\beta_R) + \lambda\, \|\beta_R + \beta_N\|_1 \tag{16}$$

where the last term in the right hand side is the lasso's ℓ_1-norm penalty. In the case of ridge regression, $\|\beta_R + \beta_N\|_2^2 = \|\beta_R\|_2^2 + \|\beta_N\|_2^2$, which upon minimisation leads immediately to $\hat{\beta}_N = 0$, regardless of the value of β_R. However, this factorisation does not hold for the last term in (16) despite the orthogonality of β_R and β_N. Therefore, for a fixed value of β_R, minimisation of the loss function (16) with respect to β_N, where β_N belongs to the null space of X, does not necessarily imply $\hat{\beta}_N = 0$.

Conclusion: The data contain no direct information about the null-space component β_N. Conditional on β_R, the value of β_N is therefore entirely determined by the ℓ_1 penalty. Nevertheless, because the estimated β_R depends on the data, the lasso selected value of β_N can depend indirectly on the data through β_R.

Inference of marker effects using the spike and slab model when the matrix of marker genotypes is rank deficient

This subsection elaborates on the subject of Bayesian learning in the case of the spike and slab model.

The first point to be made is that the lack of direct identifiability of β via the data also holds for the Bayesian spike and slab model. However, in common with the Bayesian linear genomic model where β, a priori, is assigned a $N(0, I\sigma^2_\beta)$, the spike and slab model provides an avenue of Bayesian learning through the posterior association between β and other parameters of the hierarchical model, about which the data may provide information. This source of indirect identifiability is arguably richer in the case of the spike and slab model because the number of contributing parameters is larger and this creates more routes for indirect learning. For example, consider the unknown parameter π, the proportion of markers included. Although the likelihood does not identify the m-dimensional parameter β, it may provide information about π. Updating π updates $\Pr(\delta_j = 1|y)$ which changes the posterior distribution of β_j. Thus, there is the path

$$y \to \pi \to \delta_j \to \beta_j.$$

Likewise, learning about σ^2_β changes the distribution of the non-zero effects:

$$y \to \sigma^2_\beta \to \beta_j \mid \delta_j = 1.$$

So the relatively rich parameterisation is not merely an additional identifiability burden. It also supplies additional hierarchical pathways through which information can reach β.

An important mechanism peculiar to the spike and slab model

Beyond these hierarchical pathways of Bayesian learning, the spike-and-slab model offers another mechanism for addressing rank deficiency: it assigns positive prior probability to submodels containing only a subset of the available markers.

Under the usual setup where X is rank deficient and only a very small fraction of the m marker genotypes have an effect on the phenotype, the spike-and-slab prior can favour sparse solutions among the vectors β producing the same $X\beta$. Consequently the posterior can become concentrated on one or several sparse representations of $X\beta$. In an McMC environment, each round of the Markov chain visits a particular reduced model/configuration, defined by the realised values of the vector containing the m binary indicators δ, of which a small fraction take the value 1. Within each configuration, if the resulting matrix X_M is of full column rank, the associated β_M becomes identifiable, conditional on the configuration. The McMC sampler averages parameters over the posterior uncertainty of the different configurations without explicitly enumerating the configurations or calculating their individual posterior probabilities. Indeed, for marker β_i

$$
\begin{aligned}
\mathrm{E}(\beta_i \mid y) &= \int \beta_i p(\beta_i \mid y)\, d\beta_i \\
&= \sum_k \int \beta_i p(\beta_i, M_k \mid y)\, d\beta_i \\
&= \sum_k \left[\int \beta_i p(\beta_i \mid y, M_k)\, d\beta_i \right] \Pr(M_k \mid y) \\
&= \sum_k \mathrm{E}(\beta_i \mid y, M_k) \Pr(M_k \mid y). \qquad (17)
\end{aligned}
$$

In a spike and slab model, submodels excluding marker i contribute zero to the sum, since $\beta_i = 0$ in these submodels. Consequently,

$$
\mathrm{E}(\beta_i \mid y) = \sum_{k: i \in M_k} \mathrm{E}(\beta_i \mid y, M_k) \Pr(M_k \mid y)
$$

The posterior expectation of the effect of marker i is obtained by averaging its conditional posterior expectations over all configurations that include the marker, weighting each expectation by the posterior probability of the corresponding configuration.

Now consider the McMC sampler. Once it has reached stationarity, the pairs $(\beta_i^{[t]}, M_k^{[t]})$ have the joint posterior distribution $p(\beta_i, M_k \mid y)$. Consequently, the individual $\beta_i^{[t]}$ has the marginal posterior distribution $p(\beta_i \mid y)$. Therefore the Monte Carlo estimator of the first equality in (17) is

$$
\hat{\mathrm{E}}(\beta_i \mid y) = \frac{1}{T} \sum_{t=1}^{T} \beta_i^{[t]}
$$

This marginal posterior distribution incorporates averaging over all possible submodels. Thus, model averaging is a mathematical property of the marginal posterior distribution rather than an additional calculation performed by the sampler.

Marker inclusion probabilities in the spike and slab model

The same principle of Bayesian model averaging applies to the inclusion indicators, whose posterior expectations have a particularly simple interpretation.

In the spike and slab model, the posterior inclusion probability $\Pr(\delta = 1 \mid y)$ of a marker is obtained by summing over all the configurations that contain the marker. To see this, consider the following schematic representation of the spike and slab model,

$$
\beta_i = \alpha_i \delta_i; \quad \alpha_i | \sigma_\beta^2 \sim N(0, \sigma_\beta^2);\ \delta_i \sim \text{Bernoulli}(\pi),
$$

where α_i and δ_i are independent a priori conditional on hyperparameters. A configuration M defines a particular submodel,

$$
y = X_M \beta_M + e.
$$

Within this submodel, the value of δ_j is known:

$$\delta_j(M) = \begin{cases} 1, j \in M, \\ 0, j \notin M. \end{cases}$$

Consequently,

$$\begin{aligned} \mathrm{E}(\delta_j \mid y) &= \sum_M \delta_j(M) \Pr(M \mid y) \\ &= \sum_{M: j \in M} \Pr(M \mid y), \end{aligned}$$

and since δ_j is binary,

$$\mathrm{E}(\delta_j \mid y) = \Pr(\delta_j = 1 \mid y).$$

Therefore the posterior inclusion probability of marker j is obtained by summing over all the posterior probabilities of the configurations that include marker j.

Connection with McMC

In practice, when the number of markers m is large the 2^m configurations cannot be enumerated. Instead, the sampler visits configurations $M^{[1]}, M^{[2]}, \ldots, M^{[T]}$. At iteration t,

$$\delta_j^{[t]} = \begin{cases} 1, j \in M^{[t]}, \\ 0, j \notin M^{[t]}. \end{cases}$$

Hence,

$$\hat{\Pr}(\delta_j = 1 \mid y) = \frac{1}{T} \sum_{t=1}^{T} \delta_j^{[t]}.$$

This estimates the sum of posterior probabilities over all configurations containing marker j, without explicitly computing the posterior probability of each configuration.

Conclusion: In a Bayesian spike and slab analysis, Bayesian model averaging is inherent in the marginal posterior distributions of marker effects, which account for uncertainty over the possible configurations of included markers. McMC sampling provides a computational means of estimating posterior quantities from these distributions, without explicitly enumerating the configurations or calculating their individual posterior probabilities. Many of these configurations may correspond to marker matrices of full column rank, so that the associated marker effects are identifiable conditional on the selected submodel. Thus, by favouring sparse configurations, the spike and slab model can mitigate the identifiability problem associated with rank deficiency. However, different configurations may provide equally plausible explanations of the data, and identifiability within individual configurations does not imply that the configurations themselves can be distinguished. Bayesian model averaging accounts for this uncertainty without necessarily resolving the lack of identifiability of the complete vector of marker effects through the likelihood. Nevertheless, the hierarchical structure of the spike and slab model provides additional pathways for Bayesian learning about marker effects through their posterior association with other model parameters.

An additional consideration, particularly with correlated markers, arises in the context of gene discovery. By fitting marker effects jointly, the spike and slab model can distinguish nonzero conditional effects from marginal associations induced by linkage disequilibrium. However, when several sparse configurations provide similar explanations of the data, uncertainty remains about which particular markers have nonzero effects. This uncertainty is reflected in the marginal posterior inclusion probabilities, which may be relatively small for individual markers, even when the posterior probability that at least one marker in a group has an effect is close to one. Thus, while Bayesian model averaging can be advantageous for prediction, where different configurations may yield similar predicted values, identifying individual markers with nonzero effects may remain difficult, especially in regions of strong linkage disequilibrium.

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